Sprint 6 Review Report

Project: Music Course Platform
Sprint: Sprint 6 (Week 12–13)
Date: 13.4.2026
Scrum Master: Su Wai Phyoe
Team Members: Luo Ying, Lu Liu, Su Wai Phyoe, Yicheng Chen

Sprint Goal

Extend localization support into the database layer while improving overall code quality through static code analysis and refactoring. Prepare the system for acceptance testing and update system architecture documentation.

Key Objectives

• Implement database-level localization with UTF-8 encoding support
• Design and update database schema for multilingual data handling
• Conduct statistical code review using SonarQube / SonarScanner
• Identify code complexity, duplication, and maintainability issues
• Refactor and clean up existing codebase based on analysis results
• Design acceptance test plan (functional, usability, performance testing)
• Update system architecture (ER diagram + UML class diagram)
• Generate missing i18n localization entries from resource files

Completed Tasks

• Database Localization Implementation – Completed by Yicheng Chen
Implemented database localization strategy with UTF-8 support to ensure multilingual data storage and retrieval across the system

• Database Schema Update – Completed by Yicheng Chen
Updated database structure to support multilingual fields and localization requirements

• Statistical Code Review – Completed by Su Wai Phyoe
Performed static analysis using SonarQube/SonarScanner to evaluate cyclomatic complexity, duplication, and code quality issues

• Code Clean-Up and Refactoring – Completed by Yicheng Chen
Refactored complex methods, removed duplicate logic, and improved readability and maintainability of the codebase

• Acceptance Test Plan Design – Completed by Su Wai Phyoe
Designed functional, usability, and performance test cases without execution, mapped to system requirements

• Documentation Update – Completed by Lu Liu
Updated README with localization method explanation and maintained project documentation consistency.

• Architecture Design Update – Completed by Luo Ying
Updated ER diagram (including INSTRUMENT table) and revised UML class diagram according to system changes

Team Contribution Summary

Team Member	Assigned Tasks	Time Spent (hrs)	|In class tasks
Luo Ying	ER diagram update, UML diagram update, documentation	12 hrs	Submitted
Lu Liu	README update, localization documentation	13 hrs	Submitted
Su Wai Phyoe	Code review report, acceptance test plan, sprint	13 hrs	Submitted
Yicheng Chen	Database localization, refactoring, implementation	14 hrs	Submitted

Refactoring Summary

SonarQube findings guided the following targeted refactoring (total issues reduced from 134 → 85):

Area	Change	Rule / Reason
Logging	Replaced all System.err.println and e.printStackTrace() calls across 6 classes with SLF4J/Logback (logger.error(...))	java:S106, java:S4507
Package mismatch	Moved DatabaseConnection.java from src/main/java/dao/ to src/main/java/util/ to match its package util declaration	java:S1598 (CRITICAL)
Duplicated queries	Extracted BASE_QUERY constant in TeacherProfileDAO and LearnerProfileDAO, eliminating repeated SELECT … literals across 5 methods each	java:S1192
Magic strings	Extracted repeated i18n keys (ERROR_FILL_FIELDS, INSTRUMENT_PREFIX) into private static	java:S1192

Area	Change	Rule / Reason
	final String constants in controllers and LocalizationManager	
Unused code	Removed unused BookingDAO import, field, and instantiation from TeacherDashboardController and TeacherProfileViewController	java:S1172, java:S1068
Deprecated API	Replaced new Locale("en") / new Locale("zh") / new Locale("ar") with Locale.forLanguageTag(...)	java:S2111
Test class modifiers	Removed redundant public modifier from all 11 JUnit 5 test classes	java:S5786

Challenges and Solutions

• Database Localization Complexity
Solved by implementing UTF-8 encoding and structured multilingual database design

• High Cyclomatic Complexity in Code
Resolved by splitting large methods into smaller reusable components

• Code Duplication Issues
Removed redundant logic and improved service-layer structure

• Test Design Coverage
Ensured full coverage of functional, usability, and performance test cases without execution

Conclusion

Sprint 6 successfully extended localization into the database layer and significantly improved code quality through static analysis and refactoring. The system is now more maintainable, structured, and ready for final acceptance testing.